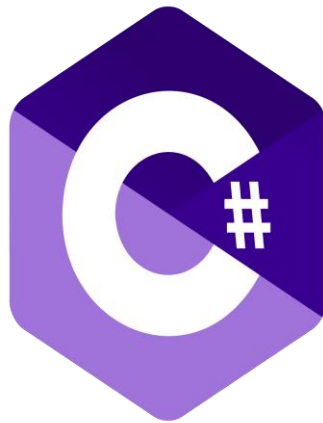


C Sharp Language



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PREPARED BY

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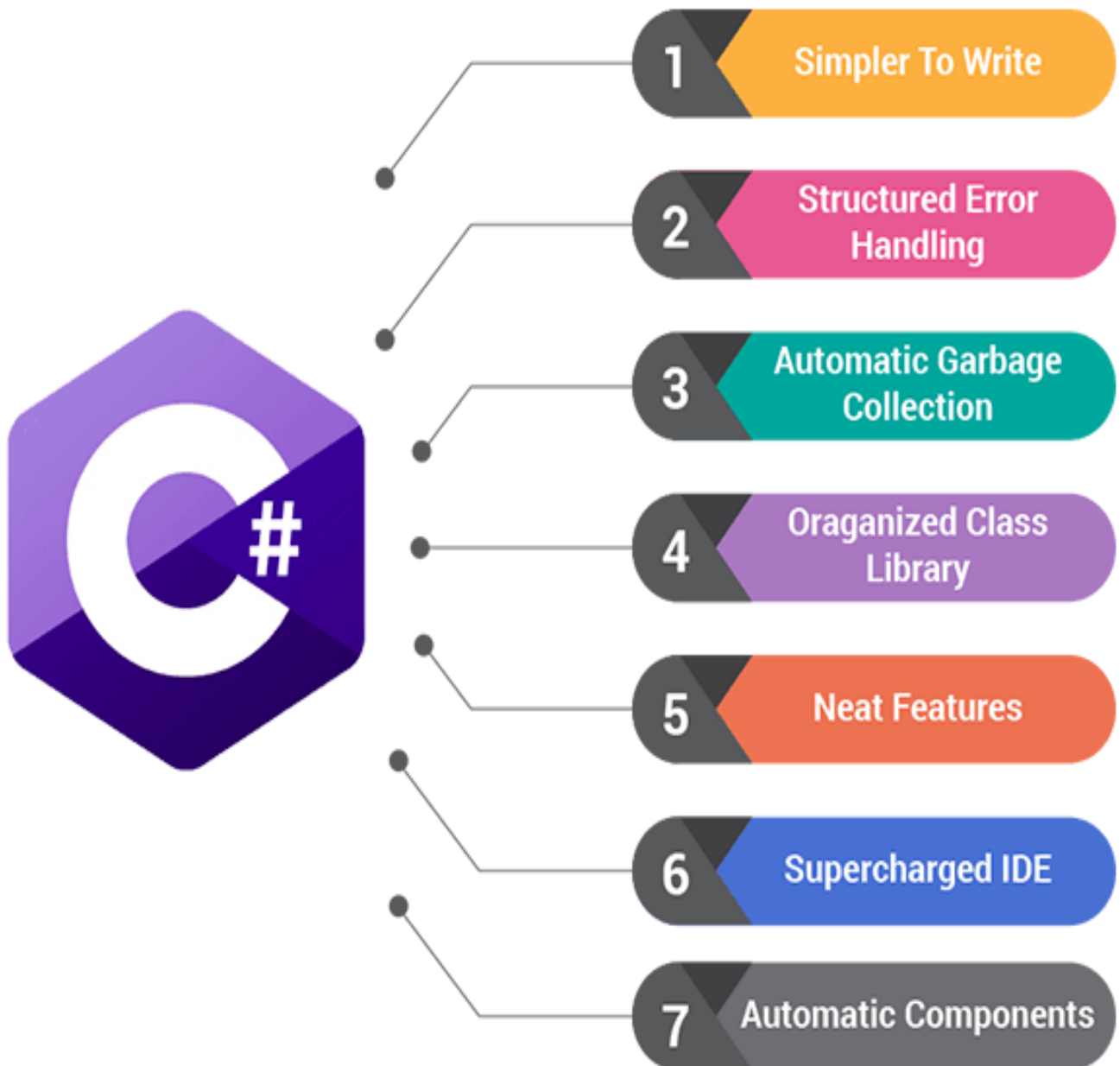
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Introduction: Characteristics of C-sharp

(1 page)

C# is a modern, high-level programming language supported by Microsoft. It combines elements of C++ and C with some features from other languages like Java. It also supports object-oriented programming concepts, like classes, objects, inheritance, and polymorphism. It's a strongly typed language, which means that every variable and expression has a type that's known at compile time so that every error will be caught early and the code will be reliable. C# is like a universal language for computers. It doesn't care if you're using a Windows PC, a Mac, or a Linux machine, It speaks their language. This makes life easier for developers because they can write code once and run it on any of these systems without any hassle. C# has a built-in cleanup system, it's a helpful system called garbage collection that takes care of memory cleanup automatically. This helps keep things organized and prevents memory problems. It's a flexible programming language that offers modern features and is commonly used for building a wide range of applications, including desktop software, web applications, and games. C# has a helpful toolbox called the .NET framework, which provides all the necessary tools for common tasks like file operations, networking, and database management. This saves developers time and effort by offering ready-to-use functionality for building software. C# is a compiled language that runs on the .NET runtime, which offers a higher speed, and the compilation and execution are very fast.



1. Question 2: Where is this programming language used?

(1 page)

1.1. Game Development

C# is a popular choice for game development, especially with the Unity game engine, which utilizes C# as its primary scripting language.

1.2. Business and Enterprise Applications

Many businesses use C# to develop internal tools, enterprise resource planning systems, and customer relationship management software.

1.3. Microsoft Office

A lot of Microsoft Office, such as Excel and Outlook are built using C sharp language

1.4. Social Media Platforms

Platforms like Facebook and LinkedIn utilize C# for user authentication, backend services, and data processing.

Question 3: How does this programming language differ from other programming languages?

C# differs from other programming languages in several ways, for example:

1.5. Syntax

C sharp has its unique syntax, but it is similar to other c style languages like C, C++, and Java, while also integrating features from languages like Delphi and Visual Basic.

1.6. Community and Adoption

C# has a large and active community of developers and is widely adopted in industries such as finance, gaming, and enterprise software development. It has a big community compared to newer or less popular languages

1.7. Strong Typing

C# uses strong typing, which means that variables must declare their data type upfront and cannot change during the program's execution. This makes C# code safer, but it might require more explicit type declarations compared to dynamically typed languages like Python.

Comparison between 3 languages:

	Python	C#	Java
Purpose	General-purpose, scripting language	General-purpose, object-oriented	General-purpose, object-oriented
Syntax	Easy to read and write	Similar to C/C++ syntax	Similar to C/C++ syntax
Typing	Dynamically-typed	Statically-typed	Statically-typed
Platform	Cross-platform (Windows, macOS, Linux)	Primarily Windows, also supported on other platforms	Cross-platform (Windows, macOS, Linux)
Development	Interpreted	Compiled	Compiled
Performance	Generally slower than compiled languages	High performance, close to compiled languages	High performance, close to compiled languages
Memory Management	Automatic garbage collection	Automatic garbage collection	Automatic garbage collection
Ecosystem	Vast collection of libraries and frameworks	Rich ecosystem with Microsoft tools and libraries	Mature ecosystem with numerous libraries and frameworks
Popularity	Very popular, widely used in various domains	Popular, widely used in Windows development	Popular, widely used in enterprise and web development
Learning Curve	Relatively easy to learn and understand	Moderate learning curve	Moderate learning curve

2. Question 4

(1 to 2 pages)

2

2.1 By default, the language is statically typed or dynamically typed?

C# is a statically typed language. Variable must have their data types declared at compile time and it cannot be changed at execution time. so when you declare a variable, you must give it a data type. For example, if you write `int myName;` it cannot change it to a different data type during the program's execution. This helps you to prevent type-related errors and ensures that variables are used consistently throughout the program.

2.2. Does the language support Object-Oriented Programming?

Yes, C# supports Object-Oriented Programming principles, such as classes and objects, so we can define a class and create objects, inheritance, polymorphism, and encapsulation. Inheritance allows you to create new classes based on existing ones, polymorphism to treat objects of different classes as objects of a common Superclass, and encapsulation which is the bundling of data and methods that operate on the data within a single unit.

2.3. Is it considered a High-Level Language?

Yes, C# is a high-level language that is easy to understand and read by humans. It offers automatic memory management, garbage collection, and a comprehensive standard library. These features simplify programming tasks, allowing developers to focus on higher-level problem-solving without getting bogged down in low-level details.

2.4. Is the language compiled or interpreted?

C sharp is a Compiled language. before being executed by the Common Language Runtime, the code written in C# is translated into an intermediate language by the compiler. This compilation process occurs before execution, to ensure better performance and optimization.

Part 2: Write a complete Java program**(1 page)**

```
package reham.projectcsci351;

/**
 *
 * @author Reham
 */
public class RehamProjectcsci351 {

    public static final int MAX = 10;

    public static void main(String[] args) {

        int listA[] = {22, 5, 18, 4, 9, 12, 13, 17, 23, 24};
        int even = 0, odd = 0;

        System.out.print("listA stores: {");

        for (int i = 0; i < listA.length; i++)
            System.out.print(listA[i] + " ");
        System.out.println("}");

        for (int i = 0; i < listA.length; i++) {
            if (listA[i] % 2 == 0)
                even++;
            else
                odd++;
        }
    }
}
```

```
    }
    int index = 0;
    while (index < listA.length) {
        System.out.println("Element at index " + index + " is: "
+ listA[index]);
        index++;
    }

    if (even == odd)
        System.out.println("listA is a balanced array");
    else
        System.out.println("listA is not a balanced array");

    for (int i = 0; i < MAX; i++) {

        int r = i;
        if (r % 2 == 0) {

            int elseVariable = i;
            System.out.println("Even number: " + r);
        } else {
            System.out.println("Odd number: " + i);
        }
    }
}
```

If I write this code using C# it will be :

```
using System;
```

```
namespace RehamProjectCSCI351
```

```
{
```

```
    class RehamProjectcsci351
```

```
    {
```

```
        public const int MAX = 10;
```

```
        static void Main(string[] args)
```

```
        {
```

```
            int[] listA = { 22, 5, 18, 4, 9, 12, 13, 17, 23, 24 };
```

```
            int even = 0, odd = 0;
```

```
            Console.Write("listA stores: {");
```

```
            foreach (int num in listA)
```

```
                Console.Write(num + " ");
```

```
            Console.WriteLine("}");
```

```
            foreach (int num in listA)
```

```
            {
```

```
                if (num % 2 == 0)
```

```
                    even++;
```

```
                else
```

```
                    odd++;
```

```
            }
```

```
            int index = 0;
```

```
        while (index < listA.Length)
        {
            Console.WriteLine("Element at index " +
index + " is: " + listA[index]);
            index++;
        }

        if (even == odd)
            Console.WriteLine("listA is a balanced
array");
        else
            Console.WriteLine("listA is not a balanced
array");

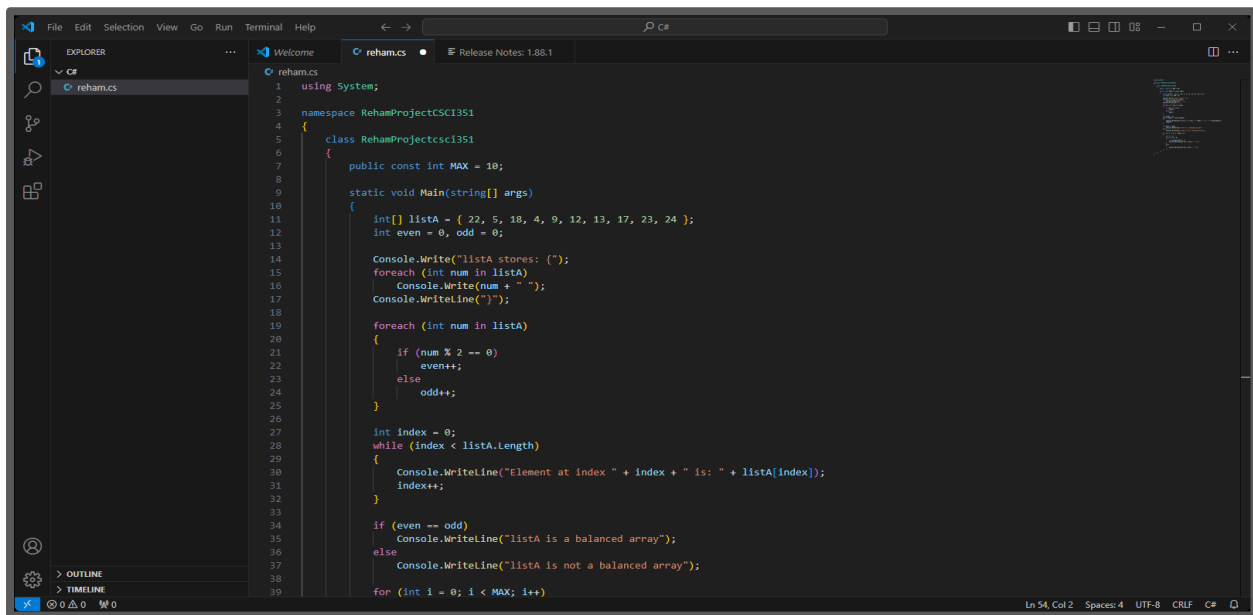
        for (int i = 0; i < MAX; i++)
        {
            int r = i;
            if (r % 2 == 0)
            {
                int elseVariable = i;
                Console.WriteLine("Even number: " + r);
            }
            else
            {
                Console.WriteLine("Odd number: " + i);
            }
        }
    }
}
```

So we conclude that :

- namespace RehamProjectCSCI351 in C# is equivalent to package reham.projectcsci351 ; in java.
- class program in C# is equivalent to public class RehamProjectcsci351 in Java.
- static void Main(string[] args) in C# is equivalent to public static void main(String[] args) in java.
- public const int MAX = 10; in C# is equivalent to public static final int MAX = 10; in java.
- Console.WriteLine(""); in C# is equivalent to System.out.println("") in java.
- if (condition) is the same in both languages.
- while (index < listA.Length) is also the same in both languages.
- for (int i = 0; i < MAX; i++) is also the same .

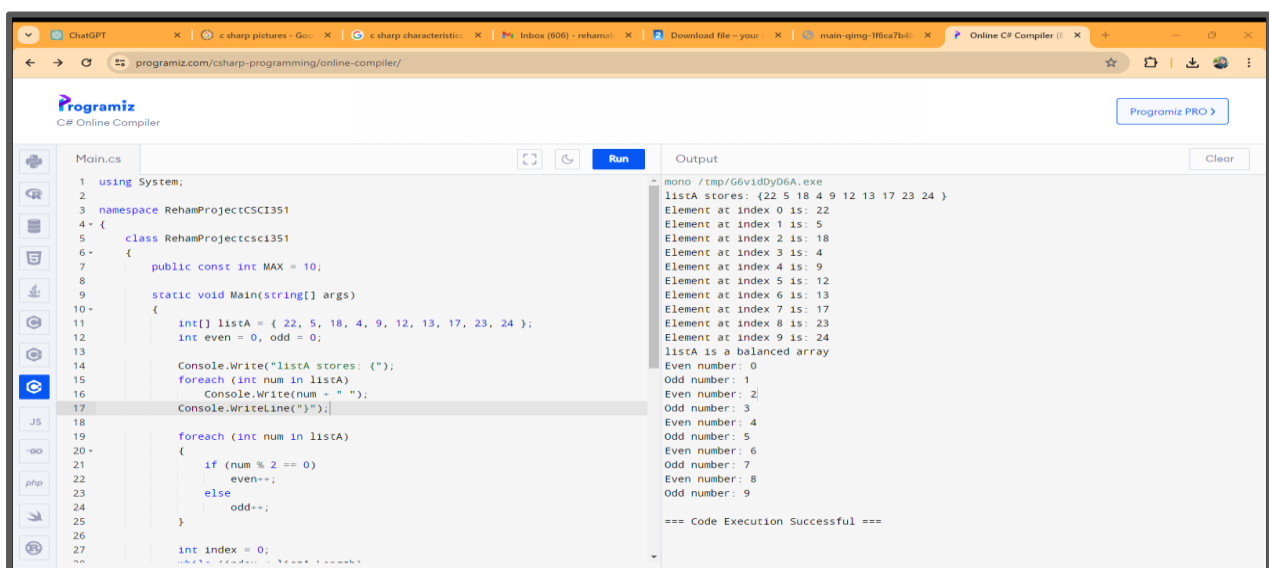
How to create the C# program?

You can create it using Visual Studio Code or other applications. Using Visual Code Studio, we create a file and put its extension .cs so it will be a C# program. or we can go to an online compiler and create it too.



The screenshot shows the Visual Studio Code editor with a C# file named 'reham.cs'. The code defines a namespace 'RehamProjectCSCI351' and a class 'RehamProjectCSCI351'. Inside the class, there is a constant 'MAX' set to 10 and a static method 'Main' that takes an array of strings 'args'. The 'Main' method initializes an integer array 'listA' with the values {22, 5, 18, 4, 9, 12, 13, 17, 23, 24}. It then iterates through the array, printing each element and its index. It also checks if the array is balanced by counting even and odd numbers. Finally, it prints the result of the balance check.

```
1 using System;
2
3 namespace RehamProjectCSCI351
4 {
5     class RehamProjectCSCI351
6     {
7         public const int MAX = 10;
8
9         static void Main(string[] args)
10        {
11            int[] listA = { 22, 5, 18, 4, 9, 12, 13, 17, 23, 24 };
12            int even = 0, odd = 0;
13
14            Console.WriteLine("listA stores: (");
15            foreach (int num in listA)
16            {
17                Console.WriteLine(num + " ");
18            }
19            Console.WriteLine(")");
20
21            foreach (int num in listA)
22            {
23                if (num % 2 == 0)
24                {
25                    even++;
26                }
27                else
28                {
29                    odd++;
30                }
31            }
32
33            int index = 0;
34            while (index < listA.Length)
35            {
36                Console.WriteLine("Element at index " + index + " is: " + listA[index]);
37                index++;
38            }
39
40            if (even == odd)
41            {
42                Console.WriteLine("listA is a balanced array");
43            }
44            else
45            {
46                Console.WriteLine("listA is not a balanced array");
47            }
48
49            for (int i = 0; i < MAX; i++)
```



The screenshot shows the Programiz Online C# Compiler. The code is identical to the one in the Visual Studio Code screenshot. The output shows the execution results, including the list of elements, their indices, and the balance check result.

```
mono /tmp/G6vidDyD6A.exe
listA stores: {22 5 18 4 9 12 13 17 23 24 }
Element at index 0 is: 22
Element at index 1 is: 5
Element at index 2 is: 18
Element at index 3 is: 4
Element at index 4 is: 9
Element at index 5 is: 12
Element at index 6 is: 13
Element at index 7 is: 17
Element at index 8 is: 23
Element at index 9 is: 24
listA is a balanced array
Even number: 0
Odd number: 1
Even number: 2
Odd number: 3
Even number: 4
Odd number: 5
Even number: 6
Odd number: 7
Even number: 8
Odd number: 9
=== Code Execution Successful ===
```

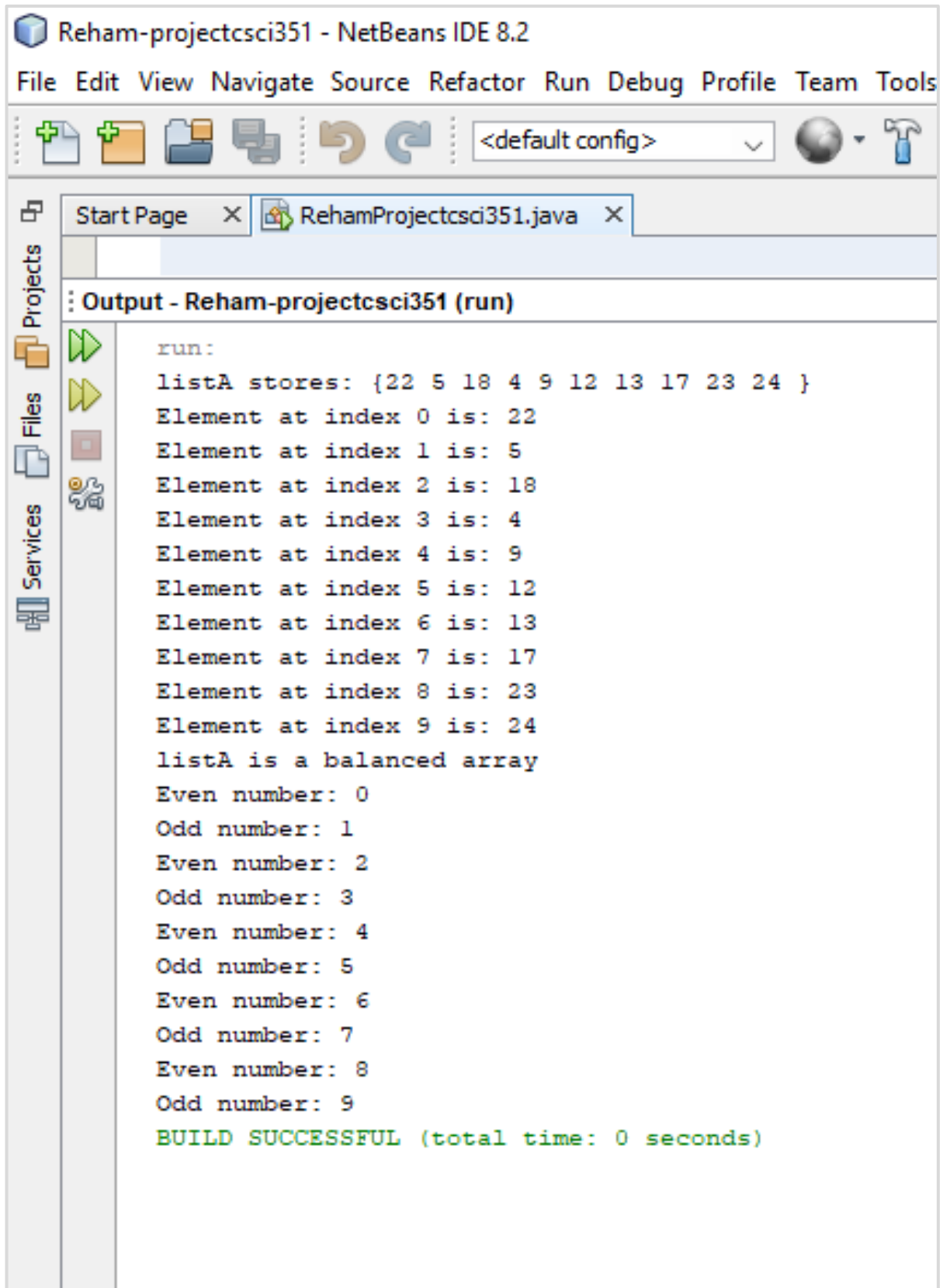
C# and Java will give us the same output:

C# Output:

```
mono /tmp/G6vidDyD6A.exe
listA stores: {22 5 18 4 9 12 13 17 23 24 }
Element at index 0 is: 22
Element at index 1 is: 5
Element at index 2 is: 18
Element at index 3 is: 4
Element at index 4 is: 9
Element at index 5 is: 12
Element at index 6 is: 13
Element at index 7 is: 17
Element at index 8 is: 23
Element at index 9 is: 24
listA is a balanced array
Even number: 0
Odd number: 1
Even number: 2
Odd number: 3
Even number: 4
Odd number: 5
Even number: 6
Odd number: 7
Even number: 8
Odd number: 9

=== Code Execution Successful ===
```

Java Output:



Reham-projectcsci351 - NetBeans IDE 8.2

File Edit View Navigate Source Refactor Run Debug Profile Team Tools

Start Page x RehamProjectcsci351.java x

Output - Reham-projectcsci351 (run)

```
run:
listA stores: {22 5 18 4 9 12 13 17 23 24 }
Element at index 0 is: 22
Element at index 1 is: 5
Element at index 2 is: 18
Element at index 3 is: 4
Element at index 4 is: 9
Element at index 5 is: 12
Element at index 6 is: 13
Element at index 7 is: 17
Element at index 8 is: 23
Element at index 9 is: 24
listA is a balanced array
Even number: 0
Odd number: 1
Even number: 2
Odd number: 3
Even number: 4
Odd number: 5
Even number: 6
Odd number: 7
Even number: 8
Odd number: 9
BUILD SUCCESSFUL (total time: 0 seconds)
```

Identify the scope(s) of each variable in your Java program.

Java program is a statically typed language, in which all variables are declared at compile time. `listA`, `even`, `odd`, and `index` are scoped to the main method and statically typed because their data types are explicitly declared at the compile time. `r` and `elseVariable` are scoped to the for loop and it's the subsequent if-else statement and also statically typed. Also, `max` is an instance variable statically typed, and it is scoped to the class. global variable is `max`, and the local variables of the main are `listA`, `even`, `odd`, and `index`. And local variables for loop is `i`, and `r`, and the else variable for loop is also.